

AGENTS.md — Fly GACA

SECTION 06-operations-it

TYPE document

STATUS

active

OWNER Founder

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How to use this file. Place it at the root of the `flygaca` repository. Google Antigravity (and other agentic tools) read `AGENTS.md` automatically as standing context for every agent run. It is the single source of truth for what Fly GACA is, what the constraints are, and the order in which it gets built.

Companion files: `flygaca-claude-briefing.md` (same plan, XML-tagged for Claude) and `fly-gaca-master-plan.docx` (the human-readable master plan).

1. Project context

Fly GACA is an **educational digital aviation platform and open regulatory library** for Saudi Arabian civil aviation. Tagline: *"The whole of Saudi aviation, in one cockpit."* It serves pilots, operators, instructors, and cadets in the Kingdom of Saudi Arabia.

Primary domain	flygaca.com (live on server 72.62.20.20)
Repository	github.com/ay2m/flygaca
Owner	Captain Adel Al-Subaie — Chief Instructor, ATPL, CFII
LinkedIn	sa.linkedin.com/in/captadel
Build mode	Solo founder · no fixed deadline · phased delivery

Problem it solves. Saudi pilots juggle scattered PDFs, paper logbooks, calculators, and many reference sites to study and stay current. The GACAR (General Authority of Civil Aviation Regulations) is rich, evolving, and spread across dozens of parts. Fly GACA fixes fragmented access, slow lookups, and the absence of GACAR-aware pilot tooling.

2. Current status — from scratch

Build from zero. Treat nothing as already built. The brand identity (Mark v2, "Falcon over Kingdom") and several business-document drafts exist; the software does not. Do not assume any feature, file, or deployment exists unless verified. Any earlier mention of a shipped "v1.0.0" is aspirational — Section 9 below is the real path.

3. Strategy

A single funnel: the **free GACAR library** is the wedge (lowest risk, highest trust, useful alone) → **Captain Adel**, the cited-source AI assistant, is the **paid premium layer** → **flight schools and operators** buying cadet seats are the revenue engine.

Captain Adel is built **once** as a single assistant service — served inside flygaca.com behind the paywall, and reachable via captadel.com as a marketing front door. One brain, two front doors, **never two codebases**. The sub-brand-vs-standalone question is deferred; the build does not depend on it.

Scope discipline: the first public launch is the library and search **only**. Everything else is post-launch, sequenced in Section 9.

4. Legal & compliance constraints — non-negotiable

These override all other considerations.

- Fly GACA is an **EDUCATIONAL** tool and is **NOT affiliated** with the General Authority of Civil Aviation. The official, authoritative source for any regulation is always GACA (gaca.gov.sa).
- No answer, feature, or copy may imply official status. Every surface must tell users to verify against the latest official GACA publication.
- The project handles personal data and must comply with **Saudi PDPL**. Respect data sovereignty in every hosting and storage decision.
- The three open risks in Section 6 must be respected — do not build as if they are settled.

5. Knowledge-base corpus policy

What Captain Adel may learn from is a legal **and** safety decision. Three tiers, fixed:

- **HOST — safe core:** GACAR parts, Saudi AIPs, GACA advisory circulars / guidance. All GACA-published. Pending the redistribution check in Section 6.
- **HOST — original:** Fly GACA's own training and exam material, authored from the Captain Adel Curriculum Map. Scraped third-party question banks are **not** permitted.
- **DO NOT HOST — cite & refer only:** ICAO Annexes / SARPs (ICAO copyright; substance already reflected in GACAR) and aircraft manuals / POH / AFM (manufacturer copyright; aircraft-specific data is a safety hazard if quoted generically). Captain Adel cites these and links to the official source; it never reproduces them and never substitutes for a POH/AFM.
- **AIP freshness:** AIPs change every 28-day AIRAC cycle. Every AIP-sourced answer must carry an effective date and the line *"not for operational use — verify the current AIP and NOTAMs."*

6. Open risks — resolve before heavy building (Phase 0 gate)

1. **Redistribution rights** for GACAR, AIPs, and advisory circulars — the launch-blocker. Confirm the Saudi position or obtain GACA's permission. Interim: source, version, and link to GACA's copy; launch as a deep-linking index if not yet cleared.
2. **The "GACA" name** leans on the government authority's identity — trademark and passing-off exposure. Needs a trademark check and an unmissable disclaimer everywhere.
3. **Domain canonicalization** — flygaca.com is owned; the brand sheet shows flygaca.sa. Assume flygaca.com unless changed.

Not legal advice — a Saudi IP lawyer should confirm items 1 and 2.

7. Hosting & infrastructure

Managed-first. Run the platform on Firebase; repurpose the VPS — it is **not** the production front door.

- **Firebase Hosting** for the static PWA; **Firebase Auth, App Check, security rules; Cloud Firestore in me-central2 (Dammam)** for data sovereignty.
- **VPS (72.62.20.20)**: hosts the Python document-ingestion / RAG-indexing pipeline, the eval harness, and staging builds. A compute box, not the live site.
- **To verify**: the VPS region, and the data-residency of AI inference (Gemini / Vertex).

8. Tech stack

Layer	Choice
Frontend	Vanilla JavaScript (ES2022), HTML5, CSS3, PWA — no framework
Backend	Firebase Cloud Functions (Node.js 20)
Database	Cloud Firestore (me-central2 region)
Auth & security	Firebase Authentication, App Check, strict security rules
AI assistant	Captain Adel — Gemini, custom system prompt + RAG pipeline
Document pipeline	Python 3.11 for chunking & indexing; Node.js builders
Hosting & CI/CD	Firebase Hosting + GitHub Actions
Observability	Web Vitals + custom error beacon
Testing	Playwright E2E; custom eval harness for the assistant

Do not introduce a frontend framework or swap a core technology without explicit approval.

9. Build plan — phases 0 to 6

Phases are **sequential by dependency, not by date**. Each ends with something usable. Do not start a phase whose predecessor's "Done when" is unmet.

Phase 0 — Foundations

Resolve redistribution rights; pick the canonical domain and lock the legal/brand name; register the legal entity (Saudi Business Center; consider Monshaat SME and NTDP); create the repo with the Section 10 structure; create the Firebase project in me-central2; set up the VPS as the pipeline box. **Done when:** the copyright question has an answer, the entity is registered, the empty repo and Firebase project exist.

Phase 1 — The library goes live

Static PWA shell on the Falcon brand; curate and text-extract the GACAR corpus; library index + document reader; client-side search over a prebuilt index; brand assets (favicon, app icon, OG card, lockups); legal pages (educational disclaimer, not-affiliated notice, terms, attributions); deploy to Firebase Hosting; point flygaca.com DNS. **Done when:** a visitor can search "night currency," open Part 91, read it on mobile, and every page makes the unofficial/educational status obvious.

Phase 2 — Captain Adel answers

Ingestion pipeline on the VPS (Python 3.11, chunk + embed); a vector store; the RAG Cloud Function (retrieve → prompt → Gemini → answer + citations); bilingual chat UI at chat.html with captadel.com as a front door; Captain Adel's system prompt (conservative, always cites, defers to GACA, refuses POH substitution); the eval harness with question sets; App Check + per-user rate limits. **Done when:** the eval set passes its quality bar, every answer carries citations, no answer asserts official status. Launch free/beta — paywall arrives in Phase 5.

Phase 3 — Pilot accounts & tools

Gated: the PDPL compliance program and DPIA must be signed off first. Firebase Auth; Firestore data model + strict per-user security rules; App Check enforced; digital logbook with GACAR currency tracking (VFR/IFR/night/landings); core calculators (weight & balance first); profile and settings; data export and full account deletion (PDPL rights). **Done when:** a pilot can register, log a flight, see currency, export a report, and fully delete their account and data.

Phase 4 — Arabic, offline & polish

Full Arabic RTL parity; offline-aware service worker (library readable offline); accessibility and Web Vitals; Playwright E2E coverage; the error beacon and observability. **Done when:** the UI works in Arabic RTL, the library works offline, performance is green.

Phase 5 — Money & flight schools

Pricing model + pricing.html; free-vs-paid split with Captain Adel behind the paywall; a Saudi payment gateway (e.g. Moyasar or Tap) wired to ZATCA Fatoora e-invoicing; subscription billing and VAT; flight-school / operator accounts (schools.html, admin view) with cadet seat management. **Done when:** a pilot can subscribe and receive a compliant e-invoice; a school can buy and manage cadet seats.

Phase 6 — Reach

Capacitor native wrapper for the iOS and Android stores; a documented public API. **Done when:** the app is published in both stores and the API has versioned endpoints.

10. Repository structure

```
flygaca/  
  assets/           Static assets (CSS, JS, character art, icons)  
  assistant/        Captain Adel – system prompt, KB scope, Gemini integration, rag.py  
  build_finance/    Internal docx/xlsx builders  
  evals/            Assistant evaluation harness and question sets  
  functions/        Firebase Cloud Functions + RAG service (functions/rag/)  
  library/PDFs/     GACAR source documents (curated corpus)  
  legal/            Notices, terms, attributions, disclaimers  
  office/           Operational tools and documents  
  tests/            Playwright E2E tests  
  .github/          Workflows, issue templates, Dependabot  
  *.html            flygaca, dashboard, library, lessons, exam, chat, pricing,  
                   schools, about, admin, settings
```

11. Business & company track

Runs alongside the build; largely finalizes drafts already in the project knowledge base.

- **With Phase 0:** register the entity (Saudi Business Center; Monshaat SME; pursue NTDP); open a business bank account.
- **With Phase 1:** finalize terms, EULA, disclaimers, and privacy notice.
- **Before Phase 3:** sign off the PDPL compliance program and DPIA — the personal-data gate.
- **With Phase 5:** ZATCA — VAT registration and Fatoora e-invoicing.
- **Go-to-market:** publish regulatory explainers from Captain Adel's LinkedIn; seed with flight schools and pilot communities; capture a waitlist from launch.

12. Brand system

From the Fly GACA Brand Identity Sheet (Mark v2, Falcon-over-Kingdom).

Token	Hex	Role
Falcon Night	#0A0E12	Primary canvas — dark-first UI
Falcon Teal	#2D6E8A	Primary — buttons, links, focus
Falcon Sage	#8FC9A8	Secondary accent, success
Falcon Gold	#C8A04A	Heritage accent — eyebrows/version stamps only, sparingly
Teal Bright	#4A9CB8	Hover, links on dark, focus rings
Ivory	#F5F2ED	Reading surface — documents & library
Falcon Deep	#0F1A24	Elevated cards
Falcon Mist	#1A2A38	Dividers, subtle borders

- **Typography:** Inter Tight throughout. Wordmark sets "Fly" solid, "GACA" teal→sage gradient.
- **The mark:** two falcon wings inside the Saudi silhouette, letter F in negative space. Never stretch, skew, rotate, recolour, or outline. Min 4.5:1 contrast, min size 16 px, preferred surface Falcon Night.
- The UI is dark-first; documents use the Ivory reading surface.

13. Agent operating rules

- Work **one phase at a time**, in order. Do not pull work forward across a "Done when".
- When a request is ambiguous, **ask before assuming** — especially on anything touching regulations, legal constraints, or the corpus policy.
- Keep all aviation and regulatory content **accurate and conservative**. Treat GACA as the sole authority; every regulatory surface reminds users to verify official sources.
- Honour Sections 4, 5, and 6 in every change. They are not optional.
- Match existing repo conventions; do not introduce a framework or swap core tech without approval.
- Write tests with the code (Playwright for UI flows, the eval harness for the assistant) and **run them** before reporting a task complete.
- Make small, reviewable commits with clear messages. Flag any security or data-sovereignty implication explicitly.
- Build for English and Arabic (RTL) from the start where the surface is user-facing.

14. Definition of done

A task is done only when: the code builds and runs; tests for the change pass; the relevant phase's "Done when" criteria are still satisfied; legal/corpus constraints are upheld; and any security or data-residency impact has been called out. Partial work, failing tests, or unverified output is **not** done.